

{ 2023 }

(1) On page no. 150



(2) On page no. 157

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(4) On page no. 181



(6) On page no. 155



(7) Littering refers to the careless or deliberate disposal of solid waste materials such as plastics, paper, glass or metals, in outdoor environments rather than appropriate waste management systems. These actions can lead to environmental pollution.



(8) On page no. 165



(9) On page no. 152



(10) On page no. 159



(11) On page no. 164



{2022}

R

Q1A (14)

UA



(2) Soil air is essential for soil fertility because it provides oxygen to plant roots, enabling them to carry out respiration and absorb nutrients more efficiently. Adequate oxygen levels in soil prevent root suffocation and promote healthy root growth. Soil air also helps in the decomposition of organic matter by providing the necessary oxygen for soil ^{micro}organisms, which release nutrients essential for plant growth.



(3) There are following control measures for water erosion:-

- Plant vegetation to stabilize soil.
- Construct dams to slow water flow in channels.
- Use blinds or geotextiles to prevent erosion.
- By promoting awareness about erosion control practices.



(6) On Page no. 162



(7) Chemical speciation refers to the process of identifying and quantifying the different forms, or species, of an element within a sample. These species can include different oxidation states, molecular forms and physical states.

This concept is crucial for the understanding of the toxicity, mobility, bioavailability and reactivity of an element can vary significantly depending on its specific chemical form. e.g. the toxicity of chromium depends on its speciation: Cr(III) is an essential nutrient at trace levels while Cr(VI) is highly toxic and carcinogenic.



(8) On page no. 180

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(11) On page no. 152



(15) clay is important in soil's nutrient-holding capacity due to its

high surface area and cation exchange capacity. These properties allow clay to retain and release essential nutrients for plant growth, improve water retention and enhance soil structure.

←→
{2021}

R

(6)*

UA

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(3) On page no. 161

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(4) Soil can hold cations because of the negatively charged surfaces of clay particles and organic matter. These negative charges attract and hold positively charged ions such as K^+ , Ca^{2+} and Mg^{2+} . This ability known as cation exchange capacity. While soil primarily hold cations, they can also adsorb anions to a lesser extent, depending on the soil's composition and pH.

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(5) Soil holds water through adhesion to soil particles and organic matter, cohesion between water molecules and capillary action in soil pores.

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(9) Two contributions of agrochemicals to soil pollution are:

- Pesticide contamination
- Nutrient imbalance

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{2020}
R (2), (3), (5), (6)

↔
(1) The toxicity of Cr is primarily associated with Cr(IV) not Cr(III), because Cr(IV) is a strong oxidizing agent, easily penetrates cells, generates harmful reactive oxygen species and is recognized as a carcinogen. In contrast, Cr(III) is less reactive, poorly absorbed by cells, essential in trace amount for human nutrition and significantly less toxic.

↔
(8) On Page no. 172
↔

{ 2019 }

R

(c), (h)



(a) See on Page no. 149 & 150



(e) See on Page no. 160



(f) See on Page no. 166



(g) Biodiversity is affected by the presence of pollution through following ways:

- chemical pollutants harm organisms, reducing population size.
- Pollutants accumulate in organisms and become more concentrated up the food chain, affecting higher trophic levels.
- Pollutants alter habitat conditions, disrupt reproductive system and give an advantage to pollution tolerant species, reducing overall biodiversity.



(h) micronutrients are beneficial for plant growth due to these

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role in following processes:

- Enzyme Activation
- Photosynthesis
- Respiration
- Nitrogen fixation
- Disease resistance
- Structural Integrity
- Protein and Hormone synthesis

←→
{2018}

R

(7)(8)

UA

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(2) On page no. 149

←→

(3) Industry contributes to soil pollution through improper disposal of wastes, accidental spills, atmospheric deposition of pollutants, mining activities, chemical manufacturing and urban development impact

These activities release contaminants such as heavy metals, chemicals and construction debris into the soil, harming ecosystems and potentially affecting human health.

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(9) On page no. 182

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{ 2017 }

UA

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(1), (2), (4), (5), (6)



(8) Humus plays a crucial role in soil conservation. It enhances soil structure by binding particles together which reduces erosion caused by water and wind.

Additionally, humus improves water retention, allowing soil to hold more moisture and reducing runoff. It also increases ^{es} nutrients and providing a favorable environment for beneficial soil organisms.

{ 2016 }

R

(2), (3), (4), (10)

{ 2015 }

R

(2), (3), (4), (9), (10)

{ 2014 }

R

(2), (4), (10), (3)



Other solvents dissolve and carry away nutrients or contaminants from the soil or another substance. This movement occurs as water percolates through the soil, carrying dissolved substance with it, which can lead to nutrient depletion in the soil or contamination of groundwater.

1 Leaching is the process by which water or →

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